Reputation-Weighted Multiplicative VCG Mechanism: Implementation Note

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Abstract

This note is a self-contained implementation guide for the reputation-weighted multiplicative VCG mechanism for multi-agent failure attribution. It provides the problem setup, the key definitions, all algorithms in pseudocode, prompt templates, hyperparameter recommendations, and the experimental protocol for the Who&When benchmark. Proofs and extended theoretical analysis are deferred to the companion theory notes.

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1 Problem Setup

Definition 1 (Trace)**.** *A failed trace of length T with M agents is $\mathrm{Tr} = \{(i_t, s_t, a_t)\}_{t=1}^{T}$, where $i_t \in [M]$ is the active agent at step t, s_t is the system state, and a_t is the action taken. The task outcome is $W = 0$ (failure).*

Definition 2 (Fix necessity)**.** *For each step $t \in [T]$, $\theta_t \in [\varepsilon, 1-\varepsilon]$ is the* true fix necessity*—the (unobservable) probability that step t requires correction. Here $\varepsilon \in (0, 1/4)$ is a fixed clipping parameter (default 10^{-6}).*

Definition 3 (Agent step sets)**.** *$\mathcal{T}_i = \{t \in [T] : i_t = i\}$ is the set of steps where agent i was active. The sets $\{\mathcal{T}_i\}_{i=1}^{M}$ partition $[T]$.*

Definition 4 (Discriminators and reports)**.** *$K \geq 2$ heterogeneous LLM discriminators $D_1, \ldots, D_K$ each evaluate every step. Discriminator D_k produces a* report *$\hat{\theta}_t^k \in [\varepsilon, 1-\varepsilon]$ for each t. Write $\hat{\boldsymbol{\theta}}^k = (\hat{\theta}_1^k, \ldots, \hat{\theta}_T^k)$.*

Remark 1 (Three levels of "θ")**.** *It is important to distinguish three quantities:*

- *$\theta_t \in [\varepsilon, 1-\varepsilon]$: the* true *fix necessity at step t (unknown, non-random).*
- *$\boldsymbol{\theta}^k = (\theta_t + \epsilon_t^k)_{t=1}^{T}$: the* true assessment *of discriminator D_k—what D_k* would *report if it were perfectly truthful. Here ϵ_t^k is D_k's idiosyncratic noise. This is also unobservable.*
- *$\hat{\theta}_t^k$: the* actual report *of D_k (the output of the LLM query). In the theory, DSIC ensures that the optimal strategy is $\hat{\theta}_t^k = \theta_t^k$, i.e., truthful reporting. In practice, we observe $\hat{\theta}_t^k$ and the OMD calibration steers it toward θ_t^k.*

For implementation purposes, only $\hat{\theta}_t^k$ is ever computed; θ_t and $\boldsymbol{\theta}^k$ appear only in the theoretical analysis.

1.1 Notation Reference

Symbol	Meaning	Symbol	Meaning
T	Steps in the trace	R	OMD calibration rounds
M	Number of agents	η_r	Learning rate at round r
K	Number of discriminators	G	Gradient bound: $2(K-1)T$
θ_t	True fix necessity (unobserved)	c_t	Fix cost threshold at step t
$\hat{\theta}_t^k$	D_k's report for step t	S^*	Blame set: $\{t : \bar{\theta}_t > c_t\}$
d_t^*	Optimal allocation at step t	Π_k	VCG payment to D_k
w_k^t	Reputation weight of D_k at step t	Π_i^{agent}	Blame payment for agent i
α_k^t	Normalised weight: $w_k^t / \sum_j w_j^t$	$\bar{\theta}_t$	Final consensus at step t
$V(\mathbf{d})$	Social welfare: $\sum_k v_k$	ε	Clipping parameter

2 The Multiplicative Value Function

2.1 Definition and Intuition

Definition 5 (Multiplicative value function)**.** *For discriminator D_k:*

$$\boxed{v_k(\mathbf{d}, \hat{\boldsymbol{\theta}}^k) \;=\; \prod_{t=1}^{T} \big[1 - (d_t - \hat{\theta}_t^k)^2\big],} \tag{1}$$

where $\mathbf{d} = (d_1, \ldots, d_T) \in [\varepsilon, 1-\varepsilon]^T$ is the fix decision vector.

Each factor $[1 - (d_t - \hat{\theta}_t^k)^2] \in (0, 1]$ measures per-step precision. The **product** structure is the key design choice: a single badly-matched step drags the entire value toward zero, coupling all steps through a serial-system logic.

Remark 2 (Why not additive?)**.** *Under the additive value function* $v_k^{\text{add}} = -\sum_t (d_t - \hat{\theta}_t^k)^2$*, the optimal allocation decomposes into a* simple per-step mean*:* $d_t^* = \frac{1}{K}\sum_k \hat{\theta}_t^k$*. Every discriminator receives equal weight regardless of quality. A single adversarial or miscalibrated LLM permanently biases* d_t^* *by* $\Theta(1/K)$*. No cross-step information is used.*

The multiplicative function creates cross-step coupling: a discriminator's influence at step t *depends on its consistency at* all other steps*, enabling automatic quality assessment.*

2.2 Reputation Weights

Definition 6 (Reputation weight)**.**

$$w_k^t(\mathbf{d}) \;:=\; \prod_{t' \neq t} \big[1 - (d_{t'} - \hat{\theta}_{t'}^k)^2\big] \;=\; \frac{v_k(\mathbf{d},\, \hat{\boldsymbol{\theta}}^k)}{1 - (d_t - \hat{\theta}_t^k)^2}. \tag{2}$$

Intuitively, w_k^t measures how consistent D_k is with the consensus $\mathbf{d}$ on every step *except* t. If D_k is an outlier at step 5, its weight at step 3 is reduced—even though the two steps are unrelated. This is the reputational feedback that is entirely absent in the additive model.

2.3 Allocation Rule

The social welfare is $V(\mathbf{d}) := \sum_{k=1}^K v_k(\mathbf{d}, \hat{\boldsymbol{\theta}}^k)$. The mechanism selects $\mathbf{d}^* = \arg\max_{\mathbf{d}} V(\mathbf{d})$.

Proposition 1 (First-order condition)**.** *Any interior maximiser satisfies, for each* $t = 1, \ldots, T$*:*

$$\boxed{d_t^* \;=\; \frac{\sum_{k=1}^K \hat{\theta}_t^k \cdot w_k^t(\mathbf{d}^*)}{\sum_{k=1}^K w_k^t(\mathbf{d}^*)}.} \tag{3}$$

This is a reputation-weighted average*:* D_k*'s report at step* t *is weighted by its cross-step consistency* w_k^t*. In the additive limit (all weights* $\to 1$*), this reduces to the simple mean.*

Remark 3 (Coupled fixed-point system)**.** *Equation* (3) *is a coupled system:* d_t^* *depends on* w_k^t*, which depends on* $d_{t'}^*$ *for* $t' \neq t$*. This is solved by Gauss–Seidel iteration (Algorithm 1). Convergence is* ***unconditional****—it works for any* T*, any* K*, any reports, with no tuning required. The proof uses* V *as a Lyapunov function: each coordinate update increases* V *(by coordinate-wise strict concavity), and* V *is bounded above.*

3 VCG Payment and Gradient

3.1 VCG Payment

Definition 7 (VCG payment)**.** *Let* $S_{-k}(\mathbf{d}) := \sum_{j \neq k} v_j(\mathbf{d}, \hat{\boldsymbol{\theta}}^j)$ *and* $\mathbf{d}_{-k}^* := \arg\max_{\mathbf{d}} S_{-k}(\mathbf{d})$*. The VCG externality payment is:*

$$\boxed{\Pi_k \;:=\; S_{-k}(\mathbf{d}_{-k}^*) - S_{-k}(\mathbf{d}^*) \;\geq\; 0.} \tag{4}$$

Π_k measures the informational contribution of D_k: how much removing D_k would hurt the other discriminators' welfare. Higher Π_k = more useful discriminator. Computing Π_k requires solving the allocation *without* D_k (a leave-one-out solve).

3.2 Gradient Structure

Proposition 2 (Gradient decomposition)**.** *The gradient of* Π_k *with respect to* $\hat{\theta}_t^k$ *decomposes as:*

$$\frac{\partial \Pi_k}{\partial \hat{\theta}_t^k} = \underbrace{\sum_{j \neq k} \frac{\partial v_j}{\partial d_t^*} \cdot \frac{\partial d_t^*}{\partial \hat{\theta}_t^k}}_{\textit{direct effect}} + \underbrace{\sum_{j \neq k} \sum_{t' \neq t} \frac{\partial v_j}{\partial d_{t'}^*} \cdot \frac{\partial d_{t'}^*}{\partial \hat{\theta}_t^k}}_{\textit{cross-step coupling (zero in additive)}} . \tag{5}$$

The cross-step term is the key: changing $\hat{\theta}_t^k$ affects d_t^*, which changes $w_j^{t'}$ for $t' \neq t$ (because $w_j^{t'}$ includes the factor at step t), which in turn changes $d_{t'}^*$. In the additive model, $d_{t'}^*$ does not depend on $\hat{\theta}_t^k$ for $t' \neq t$, so this entire term vanishes and the gradient reduces to trivial per-step averaging.

4 Key Theoretical Properties (Summary)

The following results are proved in the companion theory note. They are listed here for reference; no proofs are given.

Property	**Statement**
DSIC	Truthful reporting is a strict dominant strategy for every D_k.
Unique Nash	Truthful reporting is the unique Nash equilibrium.
Full-game eq.	The two-layer game (discriminators + agents) has a NE where discriminators report truthfully and agents minimise blame.
Unconditional convergence	Gauss–Seidel converges for *any* T, any reports. $V(\mathbf{d}^{(\ell)})$ is non-decreasing. No condition on ρ, T, or $\bar{\delta}$.
Uniqueness	Under the stochastic model, the allocation is unique w.h.p. for any T (condition on σ_H, not on T).
Output stability	Even if multiple critical points exist, the blame set, blame step, and blame agent are identical across all of them.
Regret	OMD achieves regret $O(T\sqrt{R})$.
Last-iterate conv.	With $\eta_r = O(1/r^p)$, $p \in (1/2, 1)$: $\hat{\boldsymbol{\theta}}^{(r)} \to \boldsymbol{\theta}^*$ (see Remark 7 on initialisation).
Byzantine BP	Breakdown point is $\lfloor (K-1)/2 \rfloor$. Additive model has breakdown point 0.
Minimax rate	Excess risk is $\Theta(1/\sqrt{T})$, independent of common-noise level τ^2 under equal factor loadings.
Polynomial time	Allocation and VCG payments are computable in polynomial time.

5 Algorithms

The complete pipeline consists of four phases. The data flow is:

$$\begin{array}{ccccc} \textbf{Phase 1} & \xrightarrow{\hat{\boldsymbol{\theta}}^{\text{raw}}} & \textbf{Phase 1b} & \xrightarrow{\mathbf{d}^{*,(0)},\, \hat{\boldsymbol{\theta}}^{\text{raw}}} & \textbf{Phase 2} \\ (KT \text{ LLM calls}) & & \text{(Gauss–Seidel)} & & \text{(OMD calibration)} \\ & & & \xrightarrow{\hat{\boldsymbol{\theta}}^{(R)}} & \textbf{Phase 3–4} \\ & & & & \text{(Final alloc + blame)} \end{array}$$

- Phase 1 produces raw reports $\hat{\theta}_t^{k,\text{raw}}$ ($K \times T$ matrix).
- Phase 1b computes the initial allocation $\mathbf{d}^{*,(0)}$ from the raw reports (used for diagnostics and as a baseline).
- Phase 2 takes the raw reports as OMD input and iteratively calibrates them, producing $\hat{\boldsymbol{\theta}}^{(R)}$.
- Phase 3 computes the final allocation from $\hat{\boldsymbol{\theta}}^{(R)}$ and Phase 4 attributes blame.

If Phase 2 is skipped ($R = 0$), Phase 3 simply uses $\hat{\boldsymbol{\theta}}^{\text{raw}}$ and the output of Phase 1b is the final allocation.

5.1 Algorithm 1: Gauss–Seidel Allocation Solver

Algorithm 1 SOLVEALLOCATION($\hat{\boldsymbol{\theta}}^1, \ldots, \hat{\boldsymbol{\theta}}^K;\ L,\ \tau_{\text{tol}};\ \mathbf{d}^{(\text{init})}$)

Input: Reports $\hat{\boldsymbol{\theta}}^k \in (\varepsilon, 1-\varepsilon)^T$; max sweeps L; tolerance τ_{tol}; optional warm-start $\mathbf{d}^{(\text{init})}$.
Output: Allocation $\mathbf{d}^*$, weights $\{w_k^t\}$.
Initialise: $d_t \leftarrow d_t^{(\text{init})}$ if provided, else $d_t \leftarrow \frac{1}{K}\sum_k \hat{\theta}_t^k$ for all t.
$V_{\text{old}} \leftarrow \sum_k \prod_t [1 - (d_t - \hat{\theta}_t^k)^2]$.
for $\ell = 1$ **to** L **do**
  for $t = 1$ **to** T **do**
    for $k = 1$ **to** K **do**
      $w_k^t \leftarrow \prod_{t' \neq t} [1 - (d_{t'} - \hat{\theta}_{t'}^k)^2]$ {Uses latest $d_{t'}$ (in-place)}
    end for
    $d_t \leftarrow \sum_k \hat{\theta}_t^k\, w_k^t \,/\, \sum_k w_k^t$ {Gauss–Seidel update}
    $d_t \leftarrow \text{clip}(d_t,\ \varepsilon,\ 1-\varepsilon)$
  end for
  $V_{\text{new}} \leftarrow \sum_k \prod_t [1 - (d_t - \hat{\theta}_t^k)^2]$
  Assert: $V_{\text{new}} \geq V_{\text{old}} - 10^{-12}$ {Monotonicity: must hold by theory}
  if $V_{\text{new}} - V_{\text{old}} < \tau_{\text{tol}}$ **then**
    break
  end if
  $V_{\text{old}} \leftarrow V_{\text{new}}$
end for
// Compute and store final weights for all (k, t)**:**
for $t = 1$ **to** T **do**
  for $k = 1$ **to** K **do**
    $w_k^t \leftarrow \prod_{t' \neq t} [1 - (d_{t'} - \hat{\theta}_{t'}^k)^2]$
  end for
end for
Return $\mathbf{d}$, $\{w_k^t\}_{k \in [K],\, t \in [T]}$.

Remark 4 (Critical implementation details)**.**

1. ***Gauss–Seidel, not Jacobi.*** *The weight computation $w_k^t \leftarrow \prod_{t' \neq t}[\cdots]$ uses the* latest *values of $d_{t'}$ (updated earlier in the same sweep), not the values from the previous sweep. This is essential: Gauss–Seidel converges unconditionally; undamped Jacobi may oscillate for large T.*
2. ***Monitor V, not $\|\Delta\mathbf{d}\|$.*** *The convergence criterion is $\Delta V < \tau_{\text{tol}}$, not $\|\mathbf{d}^{(\ell+1)} - \mathbf{d}^{(\ell)}\|_\infty$. The V-based criterion is theoretically justified without any contraction condition.*
3. ***V must be non-decreasing.*** *If the assert ($V_{\text{new}} \geq V_{\text{old}}$) ever fires, there is an implementation bug (most likely Jacobi-style updates or incorrect weight computation).*
4. ***Typical convergence:*** *3–15 sweeps for $T \leq 20$.*

5.2 Algorithm 2: VCG Payment Computation

Algorithm 2 COMPUTEVCGPAYMENTS($\hat{\boldsymbol{\theta}},\ \mathbf{d}^*$)

Input: Reports $\hat{\boldsymbol{\theta}}$, full allocation $\mathbf{d}^*$.
Output: Payments $\Pi_1, \ldots, \Pi_K$.
Compute $V_j^{\text{full}} \leftarrow \prod_t [1 - (d_t^* - \hat{\theta}_t^j)^2]$ for all j.
for $k = 1$ **to** K **do**
  $\mathbf{d}_{-k}^* \leftarrow$ SOLVEALLOCATION($\{\hat{\boldsymbol{\theta}}^j\}_{j \neq k};\ \text{init} = \mathbf{d}^*$) {Warm-start from $\mathbf{d}^*$}
  $V_j^{-k} \leftarrow \prod_t [1 - (d_{-k,t}^* - \hat{\theta}_t^j)^2]$ for all $j \neq k$.
  $\Pi_k \leftarrow \sum_{j \neq k} V_j^{-k} - \sum_{j \neq k} V_j^{\text{full}}$
  Assert: $\Pi_k \geq -\tau_{\text{tol}}$ {Non-negativity}
  $\Pi_k \leftarrow \max(\Pi_k,\ 0)$
end for
Return $\Pi_1, \ldots, \Pi_K$.

Remark 5. *Computing all K payments requires K leave-one-out allocation solves, each warm-started from $\mathbf{d}^*$. Total cost: $K + 1$ calls to* SOLVEALLOCATION.

5.3 Algorithm 3: Gradient via Finite Differences

Algorithm 3 COMPUTEGRADIENT($\hat{\boldsymbol{\theta}};\ k,\ t,\ \delta$)

Input: Current reports $\hat{\boldsymbol{\theta}}$; discriminator k; step t; step size δ.
Output: $g_t^k \approx \partial \Pi_k / \partial \hat{\theta}_t^k$.
$\hat{\boldsymbol{\theta}}^+ \leftarrow \hat{\boldsymbol{\theta}}$ with $\hat{\theta}_t^{k,+} = \min(\hat{\theta}_t^k + \delta,\ 1 - \varepsilon)$.
$\hat{\boldsymbol{\theta}}^- \leftarrow \hat{\boldsymbol{\theta}}$ with $\hat{\theta}_t^{k,-} = \max(\hat{\theta}_t^k - \delta,\ \varepsilon)$.
$\mathbf{d}^{*,+} \leftarrow$ SOLVEALLOCATION($\hat{\boldsymbol{\theta}}^+$) {Allocation under perturbed+ reports}
$\Pi_k^+ \leftarrow$ COMPUTEVCGPAYMENTS($\hat{\boldsymbol{\theta}}^+,\ \mathbf{d}^{*,+}$)$[k]$
$\mathbf{d}^{*,-} \leftarrow$ SOLVEALLOCATION($\hat{\boldsymbol{\theta}}^-$) {Allocation under perturbed− reports}
$\Pi_k^- \leftarrow$ COMPUTEVCGPAYMENTS($\hat{\boldsymbol{\theta}}^-,\ \mathbf{d}^{*,-}$)$[k]$
Return $g_t^k = (\Pi_k^+ - \Pi_k^-)\ /\ (\hat{\theta}_t^{k,+} - \hat{\theta}_t^{k,-})$.

Remark 6. *Each gradient evaluation for one (k, t) pair requires 2 full VCG computations. Full gradient for all (k, t): $2KT$ VCG solves. This is the computational bottleneck. Auto-differentiation (PyTorch/JAX) is strongly recommended for production use.*

5.4 Algorithm 4: OMD Calibration Update

Algorithm 4 OMDUPDATE($\hat{\theta},\ g,\ \eta$)

Input: Current report $\hat{\theta} \in (\varepsilon, 1-\varepsilon)$; gradient $g \in \mathbb{R}$; step size $\eta > 0$.
Output: Updated report $\hat{\theta}^{+}$.
$e^{+} \leftarrow \hat{\theta} \cdot \exp(\eta\, g)$
$e^{-} \leftarrow (1 - \hat{\theta}) \cdot \exp(-\eta\, g)$
$\hat{\theta}^{+} \leftarrow e^{+} \,/\, (e^{+} + e^{-})$
$\hat{\theta}^{+} \leftarrow \mathrm{clip}(\hat{\theta}^{+},\ \varepsilon,\ 1-\varepsilon)$
Return $\hat{\theta}^{+}$.

This is the standard exponentiated-gradient (mirror descent with Bernoulli entropy) update, applied coordinate-wise to each $\hat{\theta}_t^k$.

5.5 Algorithm 5: Complete VDA Pipeline

Algorithm 5 VDA: Complete Pipeline, Part 1 (Evaluation and Calibration)

Input: Failed trace $\{(i_t, s_t, a_t)\}_{t=1}^{T}$; K discriminators; OMD rounds R; LR schedule $\{\eta_r\}$; thresholds $\{c_t\}$.

// — Phase 1: Discriminator Evaluation (KT LLM calls) —

// QUERYDISC sends the prompt template (Section 6) to D_k and returns $P(\text{“A”})$ from the logit distribution.

for $t = 1$ **to** T; $k = 1$ **to** K **do**

  $\hat{\theta}_t^{k,\text{raw}} \leftarrow \text{clip}\big(\text{QUERYDISC}(D_k, s_t, a_t),\ \varepsilon,\ 1-\varepsilon\big)$

end for

// — Phase 1b: Initial Allocation (Gauss–Seidel, no LLM calls) —

$\mathbf{d}^{*,(0)},\ \{w_k^t\} \leftarrow \text{SOLVEALLOCATION}(\hat{\boldsymbol{\theta}}^{\text{raw}})$

// — Phase 1c: Pre-OMD Diagnostics —

Compute effective variances $\hat{\sigma}_k^{2,\text{eff}} = \frac{1}{T}\sum_t (d_t^{*,(0)} - \hat{\theta}_t^{k,\text{raw}})^2$ for each k.

Flag Byzantine candidates: any k with $\frac{1}{T}\sum_t w_k^t < 0.01$.

// — Phase 2: OMD Calibration (no LLM calls) —

Initialise from raw reports: $\hat{\theta}_t^{k,(0)} \leftarrow \hat{\theta}_t^{k,\text{raw}}$ for all k, t. {Preserves discriminator diversity}

for $r = 0$ **to** $R - 1$ **do**

  $\eta_r \leftarrow \eta_0/(r+1)^p$

  $\mathbf{d}^{*,(r)} \leftarrow \text{SOLVEALLOCATION}(\hat{\boldsymbol{\theta}}^{(r)})$ {At r=0: same as Phase 1b; can reuse}

  $\Pi_1^{(r)}, \ldots, \Pi_K^{(r)} \leftarrow \text{COMPUTEVCGPAYMENTS}(\hat{\boldsymbol{\theta}}^{(r)},\ \mathbf{d}^{*,(r)})$

  for $k = 1$ **to** K; $t = 1$ **to** T **do**

    $g_t^{k,(r)} \leftarrow \text{COMPUTEGRADIENT}(\hat{\boldsymbol{\theta}}^{(r)};\ k,\ t,\ \delta)$

    $\hat{\theta}_t^{k,(r+1)} \leftarrow \text{OMDUPDATE}(\hat{\theta}_t^{k,(r)},\ g_t^{k,(r)},\ \eta_r)$

  end for

  if $\|\hat{\boldsymbol{\theta}}^{(r+1)} - \hat{\boldsymbol{\theta}}^{(r)}\|_\infty < 10^{-5}$ **then**

    break

  end if

end for

Algorithm 6 VDA: Complete Pipeline, Part 2 (Aggregation and Blame)

// — Phase 3: Final Aggregation —
$\mathbf{d}^{*,\text{final}} \leftarrow \textsc{SolveAllocation}(\hat{\boldsymbol{\theta}}^{(R)})$
$\bar{\theta}_t \leftarrow d_t^{*,\text{final}}$ for all t
$\Pi_k^{\text{total}} \leftarrow \sum_{r=0}^{R-1} \Pi_k^{(r)}$ for all k

// — Phase 4: Blame Attribution —
$S^* \leftarrow \{t : \bar{\theta}_t > c_t\}$
for $i = 1$ **to** M **do**
  $\Pi_i^{\text{agent}} \leftarrow \sum_{t \in S^* \cap \mathcal{T}_i} \bar{\theta}_t$
end for
$\hat{i}^* \leftarrow \arg\max_i \Pi_i^{\text{agent}}$ {Predicted blame agent}
$\hat{t}^* \leftarrow \arg\max_t \bar{\theta}_t$ {Predicted blame step}

Return $\{\Pi_i^{\text{agent}}\}$, $\{\Pi_k^{\text{total}}\}$, $\hat{i}^*$, $\hat{t}^*$, $\bar{\boldsymbol{\theta}}$.

Remark 7 (Why initialise from raw reports, not consensus)**.** *One might consider initialising Phase 2 from the Phase 1b consensus: $\hat{\theta}_t^{k,(0)} \leftarrow d_t^{*,(0)}$ for all k, t. This is the "warm start" described in the companion theory note. However, this creates a* ***degenerate initialisation****: when all K discriminators report the same value at every step, the leave-one-out allocation $\mathbf{d}^*_{-k}$ equals $\mathbf{d}^*$ (removing one identical report does not change the weighted average), so all VCG payments $\Pi_k = 0$ and all finite-difference gradients $g_t^k = 0$. The OMD iteration never moves.*

By initialising from the raw reports *$\hat{\theta}_t^{k,\text{raw}}$, the discriminators retain their distinct reports. The VCG mechanism can distinguish their contributions, yielding non-trivial payments and gradients. The raw reports are already "close to truth" (since the LLMs are informative), so the OMD iteration starts in a reasonable region without requiring the consensus initialisation.*

If OMD calibration is not desired (e.g., for a quick first experiment), one may skip Phase 2 entirely and proceed directly from Phase 1b to Phase 3 using the raw reports: $\bar{\theta}_t \leftarrow d_t^{*,(0)}$.

6 Prompt Template

```
You are evaluating a step in a multi-agent task execution that ultimately
FAILED.

[TASK] {task_description}

[TRACE CONTEXT (steps 0 to {t})]
--- Step 0 ({history[0]["name"]}) ---
{history[0]["content"][:500]}
...
--- Step {t} ({history[t]["name"]}) ---
{history[t]["content"]}

[TASK OUTCOME: FAILED. Ground truth:  {ground_truth}]

Question:  Does the action at step {t} (by agent ``{history[t]["name"]}'')
contain an error that contributed to the task failure?

Answer (A) Yes or (B) No.
```

Extract $\hat{\theta}_t^k = P(\text{"A"})$ from the LLM's logit distribution. Truncate context to ∼500 tokens per preceding step; keep the current step t untruncated.

7 Hyperparameters

Parameter	Symbol	Default	Notes
Discriminators	K	3	3–5 diverse models; $K \geq 5$ for Byzantine robustness
OMD iterations	R	15	10–15 usually sufficient; $R = 0$ skips OMD
LR schedule	η_r	$0.3/(r{+}1)^{0.7}$	$p \in (0.5, 1)$
OMD init	—	Raw reports (Phase 1)	NOT consensus (see Remark 7)
Solver type	—	Gauss–Seidel	NOT Jacobi
Solver max sweeps	L	100	
Solver convergence	τ_{tol}	10^{-10}	Based on ΔV, not $\|\Delta \mathbf{d}\|$
Finite-diff step	δ	10^{-4}	Too small $\rightarrow$ numerical noise
Clipping	ε	10^{-6}	Prevents $\hat{\theta} \in \{0, 1\}$
Fix threshold	c_t	0.5	Application-dependent
Byzantine flag	—	mean $w_k < 0.01$	

8 Diagnostics Checklist

1. **V-trajectory:** Plot $V^{(\ell)}$ for each SOLVEALLOCATION call. Must be non-decreasing. If V ever decreases $\Rightarrow$ implementation bug.
2. **OMD convergence:** Plot $\|\hat{\boldsymbol{\theta}}^{(r)} - \hat{\boldsymbol{\theta}}^{(r-1)}\|_\infty$ vs. r. Should decrease.
3. **Reputation distribution:** Examine w_k^t per discriminator. Uniformly low weights $\Rightarrow$ poor quality.
4. **VCG payments:** $\Pi_k \geq 0$ for all k. If violated, increase solver precision.
5. **Effective-variance ranking:** $\hat{\sigma}_k^{2,\text{eff}} = \frac{1}{T}\sum_t (d_t^* - \hat{\theta}_t^k)^2$, where d_t^* and $\hat{\theta}_t^k$ are from the *same* phase (Phase 1b for pre-OMD, Phase 3 for post-OMD). Lower = better discriminator.
6. **Sensitivity:** Re-run with one discriminator removed; check that blame prediction is stable.

9 Computational Cost

Phase	LLM calls	Arithmetic
Phase 1 (evaluation)	KT	—
Phase 1b (allocation)	0	$O(K^2T^3 \log(1/\tau))$
Phase 2 (OMD calibration)	0	$O(RK^3T^3 \log(1/\tau))$
Phase 3 (final aggregation)	0	$O(K^2T^3 \log(1/\tau))$
Phase 4 (blame)	0	$O(MT)$
Total LLM calls	$\boldsymbol{KT}$	

With $K = 3, T = 10$: **30 short LLM calls**. All arithmetic completes in < 1 second on modern hardware.

10 Worked Example

Setup. $T = 4$, $M = 2$, $K = 3$. $\mathcal{T}_1 = \{1, 3\}$, $\mathcal{T}_2 = \{2, 4\}$. $W = 0$.

Phase 1: Initial reports.

	$t=1$	$t=2$	$t=3$	$t=4$
D_1	0.20	0.85	0.10	0.75
D_2	0.30	0.60	0.80	0.70
D_3	0.25	0.90	0.15	0.80
Simple mean	0.250	0.783	0.350	0.750

D_2 is an outlier at $t = 3$ (reports 0.80 vs. ∼0.12 from D_1, D_3) and at $t = 2$ (reports 0.60 vs. ∼0.87).

Phase 1b: Gauss–Seidel allocation. Starting from the simple mean. At step $t = 3$, the reputation weights reflect D_2's inconsistency at $t = 2$:

$$w_1^3 = [1-(0.250-0.20)^2][1-(0.783-0.85)^2][1-(0.750-0.75)^2] \approx 0.993,$$
$$w_2^3 = [1-(0.250-0.30)^2][1-(0.783-0.60)^2][1-(0.750-0.70)^2] \approx 0.962,$$
$$w_3^3 \approx 0.984.$$

D_2's weight at $t = 3$ is the lowest (0.962) because of its disagreement at $t = 2$. After convergence:

	$t=1$	$t=2$	$t=3$	$t=4$
Simple mean (additive)	0.250	0.783	0.350	0.750
Rep-weighted (mult.)	0.248	0.797	0.320	0.755

At $t = 3$: D_2's outlier (0.80) is down-weighted, shifting $0.350 \to 0.320$. At $t = 2$: D_2's low report (0.60) is down-weighted, shifting $0.783 \to 0.797$.

VCG payments. $\Pi_1 \approx 0.0012$, $\Pi_2 \approx 0.0003$, $\Pi_3 \approx 0.0015$. D_2 receives the lowest payment (least consistent).

Phase 4: Blame. $\bar{\theta} \approx (0.248,\ 0.797,\ 0.320,\ 0.755)$. With $c_t = 0.5$: $S^* = \{2, 4\}$. $\Pi_1^{\text{agent}} = 0$ (steps $\{1, 3\}$ below threshold). $\Pi_2^{\text{agent}} = 0.797 + 0.755 = 1.552$. **Agent 2 bears all the blame.**

11 Who&When Benchmark Protocol

11.1 Dataset

Who&When (Zhang et al., 2025) contains 184 annotated failure traces: Algorithm-Generated (126) and Hand-Crafted (58).

11.2 Data Mapping

VDA		**Who&When**
T	$\leftarrow$	`len(history)`
i_t	$\leftarrow$	`history[`t`]["name"]`
a_t	$\leftarrow$	`history[`t`]["content"]`
s_t	$\leftarrow$	`history[0:`t`]`
W	$\leftarrow$	0
Ground-truth agent		`mistake_agent`
Ground-truth step		`mistake_step`

11.3 Metrics

$$\mathrm{Acc}_{\mathrm{agent}} = \frac{1}{|\mathcal{D}|} \sum_d \mathbb{I}[\hat{i}^*_d = \texttt{mistake_agent}_d], \tag{6}$$

$$\mathrm{Acc}_{\mathrm{step}} = \frac{1}{|\mathcal{D}|} \sum_d \mathbb{I}[\hat{t}^*_d = \texttt{mistake_step}_d], \tag{7}$$

$$\mathrm{Acc}_{\mathrm{joint}} = \frac{1}{|\mathcal{D}|} \sum_d \mathbb{I}[\hat{i}^*_d = \texttt{mistake_agent}_d \ \wedge \ \hat{t}^*_d = \texttt{mistake_step}_d]. \tag{8}$$

11.4 Configuration

$K = 3$ (GPT-4o, Claude-Sonnet, Gemini-Pro), $R = 15$, $\eta_r = 0.3/(r{+}1)^{0.7}$, $c_t = 0.5$. Start with Hand-Crafted (58 traces). Cost: $3 \times 7 \times 184 \approx 3{,}864$ calls.

Baselines (best single-model, GPT-4o): 53.5% agent, 14.2% step.

11.5 Ablations

1. Additive (simple mean) vs. multiplicative (reputation-weighted).
2. $K \in \{1, 2, 3, 5\}$.
3. $R \in \{0, 5, 10, 15\}$ (where $R = 0$ skips OMD entirely).
4. Gauss–Seidel vs. Jacobi convergence comparison.